Enter to Learn, Leave to Succeed



Submitted to: Dr.Sunita Dianel

Submitted by:

Swati Bahety	**25PGDM047**
Nitesh Kumar Chourasia	**25PGDM033**
Dia Sharma	**25PGDM018**
Labhanshi Sharma	**25PGDM029**
Shivam kumar	**25PGDM044**

Acknowledgement

We would like to express our sincere gratitude to our faculty member, Dr. Sunita Daniel, for her valuable guidance, support, and encouragement throughout the development of this capstone project on *Loan Default Prediction Using Machine Learning*. Her insights, academic direction, and continuous motivation greatly helped us understand the practical application of machine learning techniques in solving real-world financial problems.

We would also like to extend our appreciation to the industry professional, Mr. Anand Modi (Solution Manager, ICICI Bank), whose interaction and insights helped us understand the practical challenges faced by financial institutions in evaluating borrower risk and loan repayment capacity. His perspective provided important industry context for identifying the problem addressed in this project.

This project provided us with an opportunity to apply important concepts of data preprocessing, exploratory data analysis, machine learning modeling, and model evaluation using techniques such as Logistic Regression, Decision Tree, and Random Forest algorithms. Through this analysis, we were able to understand how machine learning models can support banks in predicting loan default risk and improving credit risk management decisions.

Finally, we would like to thank our institution for providing us with the resources and learning environment that enabled us to explore practical applications of data science and machine learning in the field of finance. This project has significantly enhanced our understanding of how analytical tools can be used to support data-driven decision making in the banking industry.

Enter to Learn, Leave to Succeed

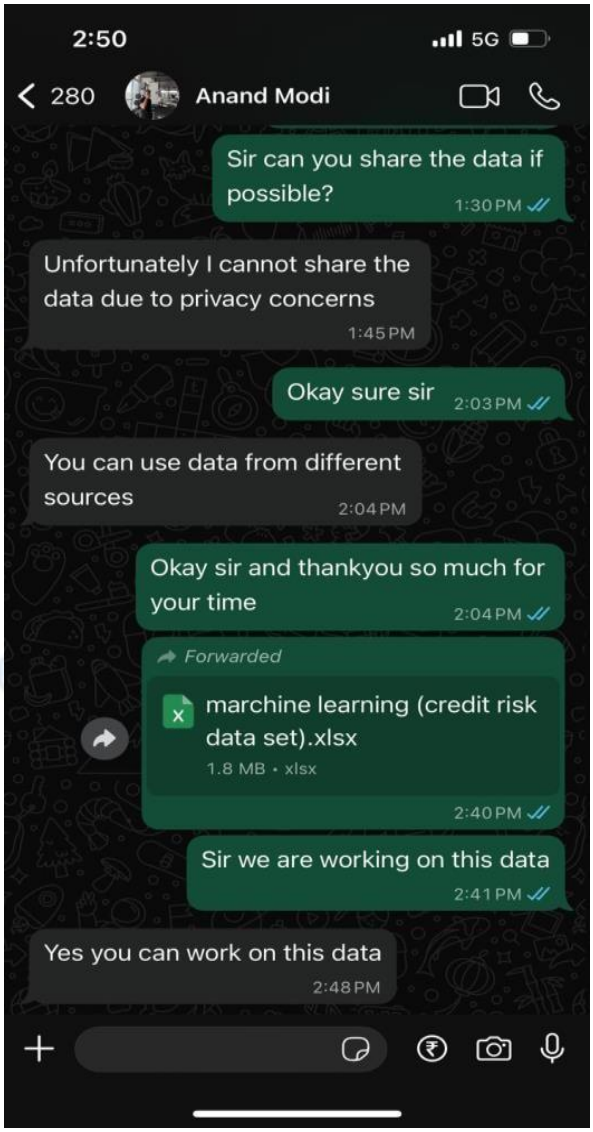
TABLE OF CONTENTS

1. **Acknowledgement**
2. **Capstone Phase 1 – Industry Problem Identification**
   - 2.1 Proof of Industry Interaction
   - 2.2 Industry Professional Details
   - 2.3 Functional Area
   - 2.4 Working Problem Identified
3. **Capstone Phase 2 – Building on Further Project**
   - 3.1 Executive Summary
   - 3.2 Introduction
   - 3.3 Problem Statement
   - 3.4 Objectives of the Project
   - 3.5 Research Methodology
   - 3.6 Dataset Description
   - 3.7 Data Pre-processing
   - 3.8 Data Analysis and Interpretation
     - 3.8.1 Target Variable Distribution Analysis
     - 3.8.2 Income and Loan Default Relationship
     - 3.8.3 Home Ownership and Loan Default
     - 3.8.4 Interest Rate and Default Risk
     - 3.8.5 Credit History Length Analysis
     - 3.8.6 Loan-to-Income Ratio Analysis
4. **Capstone Phase 3 – Model Development and Business Insights**
   - 4.1 Model Evaluation Analysis and Interpretation
   - 4.2 Model Performance Comparison
   - 4.3 Interpretation of Model Comparison
   - 4.4 Practical Implication
   - 4.5 Confusion Matrix Interpretation
   - 4.6 ROC Curve Analysis
   - 4.7 Overall Interpretation of Findings
   - 4.8 Feature Importance Analysis
   - 4.9 Feature–Variable Mapping Table
   - 4.10 Key Insights from Loan Default Prediction Analysis
5. **Limitations of the Study**
6. **Future Scope**
7. **Conclusion**

Capstone 1 – Industry Problem Identification

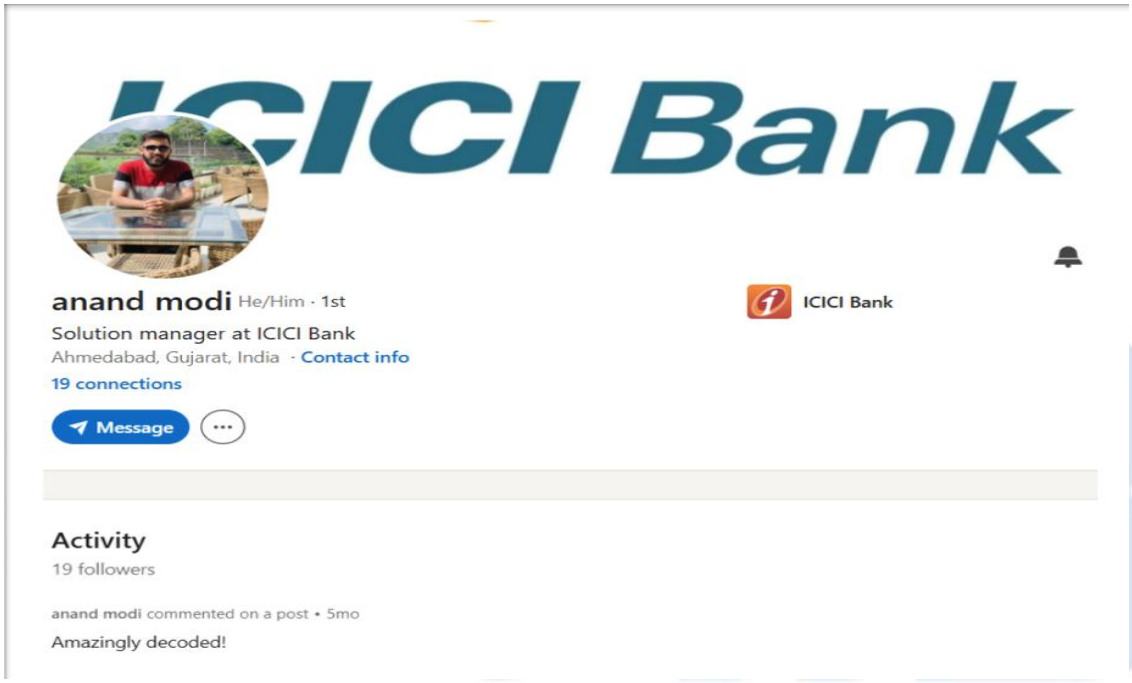
1. Proof of Industry Interaction





2. Industry Professional Details

Name: Anand Modi
Organization: ICICI Bank
Role: Solution Manager
Mode of Interaction: Personal Discussion / WhatsApp Interaction
Date of Interaction: 6 March 2026



Enter to Learn, Leave to Succeed

3. Functional Area

Working Capital Financing

Working capital financing refers to short-term credit facilities provided by banks to businesses to manage their day-to-day operational expenses such as inventory purchases, supplier payments, and operational cash flow requirements. Banks evaluate multiple financial indicators such as company revenue, credit history, cash flow stability, and repayment capacity before approving working capital loans.

4. Working Problem Identified

During interaction with a professional working in the working capital financing segment at ICICI Bank, it was observed that **evaluating the repayment capacity of businesses applying for working capital loans is a key challenge for banks**. Financial institutions receive numerous loan applications from small and medium enterprises (SMEs) seeking funds to support their operational activities.

Banks typically evaluate borrower financial indicators such as business revenue, cash flow stability, credit history, and existing debt obligations when assessing the repayment capacity of businesses.

However, accurately identifying businesses that may face repayment difficulties remains complex. Factors such as fluctuating revenues, unstable cash flows, and limited financial transparency make it difficult to assess borrower risk effectively. Loan defaults in working capital financing can increase financial losses and contribute to the rise of non-performing assets (NPAs) for banks.

Therefore, there is a need for analytical approaches that can help predict the likelihood of loan default using borrower financial and behavioral characteristics. Machine learning techniques can assist banks in analyzing borrower data and identifying patterns associated with higher default risk, enabling better credit evaluation and improved lending decisions.

Data Source and Industry Data Availability

During the industry interaction with the professional from ICICI Bank, an attempt was made to obtain real banking data related to borrower credit profiles and loan repayment behavior. However, due to strict data privacy regulations, confidentiality policies, and customer data protection norms followed by financial institutions, the organization was unable to share internal borrower data for academic use. 8 | Page Therefore, to conduct the analysis while maintaining the relevance of the study, a publicly available Credit Risk Dataset from Kaggle was used. This dataset closely matches the requirements of the project as it contains borrower financial attributes such as income, loan amount, interest rate, credit history length, employment length, and previous default records. Using this dataset allowed the project to simulate a real-world credit risk evaluation scenario and apply machine learning techniques

Excel link

marchine learning (credit risk data set).xlsx

CAPSTONE PHASE-2(BUILDING ON FURTHER PROJECT)

1.Executive Summary

This project focuses on predicting loan default risk using machine learning to support better credit decision-making in the banking sector. Based on industry interaction with a professional from ICICI Bank, borrower repayment assessment was identified as a major challenge in loan approval, particularly in reducing credit risk and non-performing assets.A credit risk dataset containing approximately 32,581 borrower records was used to analyze financial and demographic factors influencing loan repayment

behavior. Key variables included borrower income, loan amount, interest rate, employment length, credit history, and previous default records. After data preprocessing and exploratory analysis, three machine learning models were developed: Logistic Regression, Decision Tree, and Random Forest.

Among all models, Random Forest delivered the best performance with 89% accuracy and an ROC-AUC score of 0.932, indicating strong predictive capability. Feature importance analysis showed that loan-to-income ratio, borrower income, and interest rate are the most influential factors affecting default risk.The findings demonstrate that machine learning can significantly improve credit risk assessment by enabling banks to identify high-risk borrowers more accurately and support data-driven lending decisions

2. Introduction

Financial institutions play an important role in providing loans to individuals and businesses. However, lending always involves the risk that some borrowers may fail to repay their loans. Loan defaults can result in financial losses for banks and contribute to the growth of non-performing assets (NPAs).

With the increasing number of loan applications, traditional manual credit evaluation methods may not always be sufficient to accurately assess borrower risk. Machine learning techniques provide an opportunity to analyze large volumes of borrower data and identify patterns associated with loan repayment behavior.

The objective of this project is to develop a machine learning model that can predict whether a borrower is likely to default on a loan based on financial and demographic characteristics. By identifying high-risk borrowers in advance, financial institutions can make more informed lending decisions and improve their credit risk management processes.

3. Problem Statement

Banks receive thousands of loan applications from individuals and businesses seeking financial support. However, not all borrowers successfully repay their loans. Accurately predicting the likelihood of loan default is a major challenge for financial institutions.

The goal of this project is to build a machine learning model that predicts loan default risk using borrower attributes such as income, employment length, credit history, loan amount, and interest rate.

This predictive model can help banks reduce financial losses by identifying borrowers who are more likely to default.

4. Objectives of the Project

The main objectives of this project are:

- To explore and analyze borrower financial data using exploratory data analysis techniques.
- To preprocess the dataset and prepare it for machine learning modeling.
- To develop multiple machine learning models to predict loan default risk.
- To evaluate and compare model performance using various evaluation metrics.
- To identify the most important factors influencing loan default behavior.
- To generate insights that can assist banks in improving credit risk assessment.

5.Research Methodology

This project followed a structured machine learning approach to predict loan default risk using borrower financial and credit-related data. A credit risk dataset containing approximately 32,581 borrower records and 12 variables was used as the foundation for analysis. The process began with data cleaning, where missing values in important variables such as employment length and interest rate were identified and removed to improve data quality.

After preprocessing, the dataset was divided into input features and the target variable, with loan_status used to classify borrowers as default or non-default. Numerical variables were scaled, and categorical variables were encoded through a preprocessing pipeline to prepare the data for model training. Scaling was applied to ensure that numerical variables contribute equally to the model, while encoding was used to convert categorical variables into a machine-readable format. A pipeline-based approach was adopted to ensure consistency, reproducibility, and to prevent data leakage during model training.The dataset was then split into 80% training data and 20% testing data to evaluate model performance on unseen observations.

To build predictive capability, three machine learning models were applied: Logistic Regression, Decision Tree, and Random Forest. Their performance was compared using accuracy, confusion matrix, precision, recall, F1-score, and ROC-AUC, with Random Forest emerging as the strongest model for predicting borrower default risk.

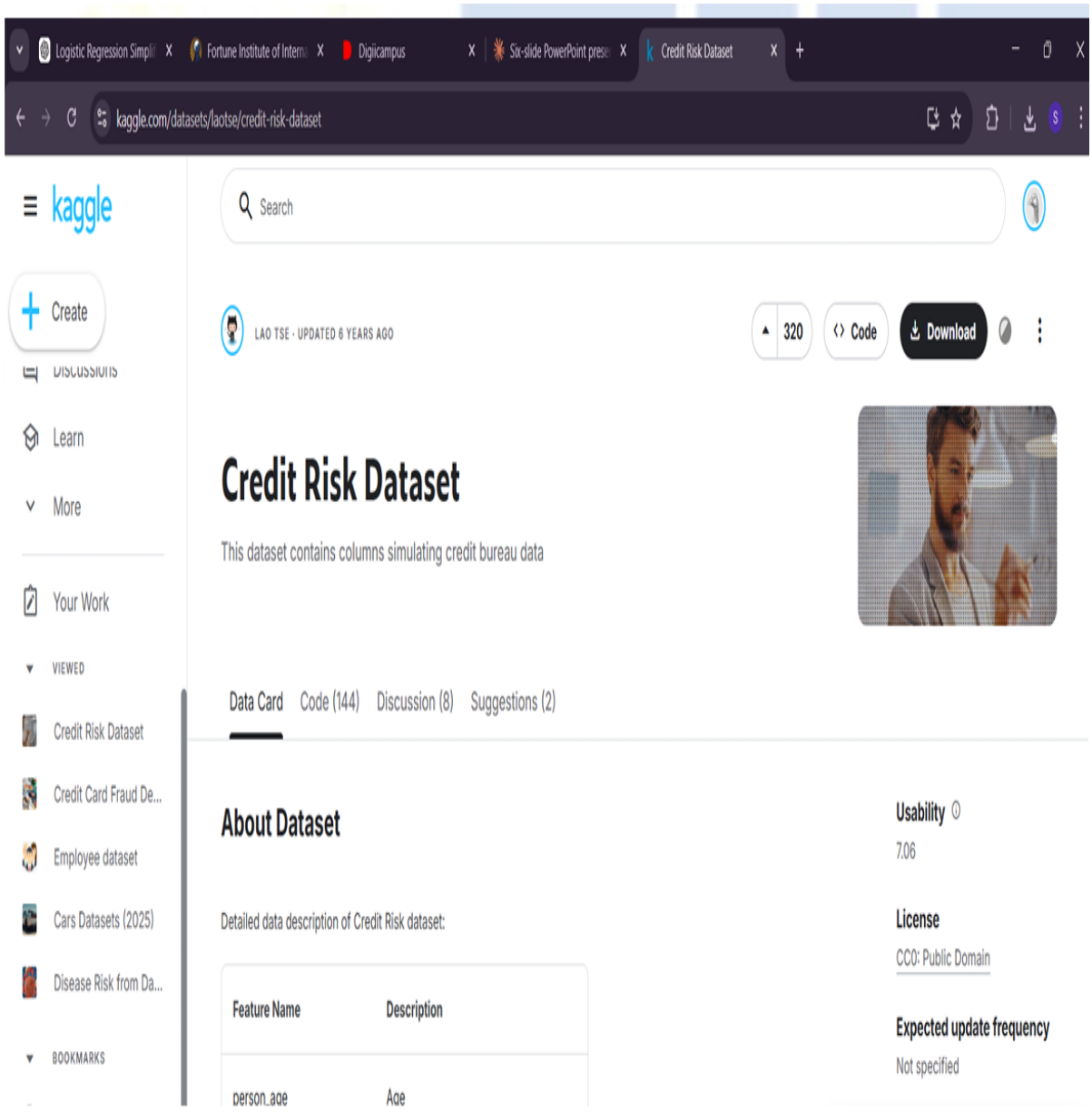
6. Dataset Description

The project uses a credit risk dataset containing borrower information and loan characteristics. The dataset includes several variables representing the financial capacity and credit behavior of borrowers.

Dataset Source: Kaggle Credit Risk Dataset -https://www.kaggle.com/datasets/laotse/credit-

risk-dataset

Snapshot Dataset Source: Kaggle Credit Risk Dataset



Total Records: ~32,581

borrower loan applications Variables: 12 borrower and

loan attributes Target Variable: loan_status (0 = repaid, 1 = default)

Unit of Analysis: Individual borrower loan application

Key variables in the dataset include:

Variable	Description
person_age	Age of the borrower
person_income	Annual income of the borrower
person_home_ownership	Housing status of the borrower
person_emp_length	Employment length
loan_amnt	Loan amount requested
loan_int_rate	Interest rate on the loan
loan_percent_income	Loan amount as percentage of income
loan_intent	Purpose of the loan
cb_person_default_on_file	Previous default record
cb_person_cred_hist_length	Length of credit history
loan_status	Target variable (loan repaid or defaulted)

The target variable **loan_status** indicates whether the borrower successfully repaid the loan or defaulted.

7. Data Preprocessing

Before building machine learning models, the dataset was cleaned and prepared.

The following preprocessing steps were performed:

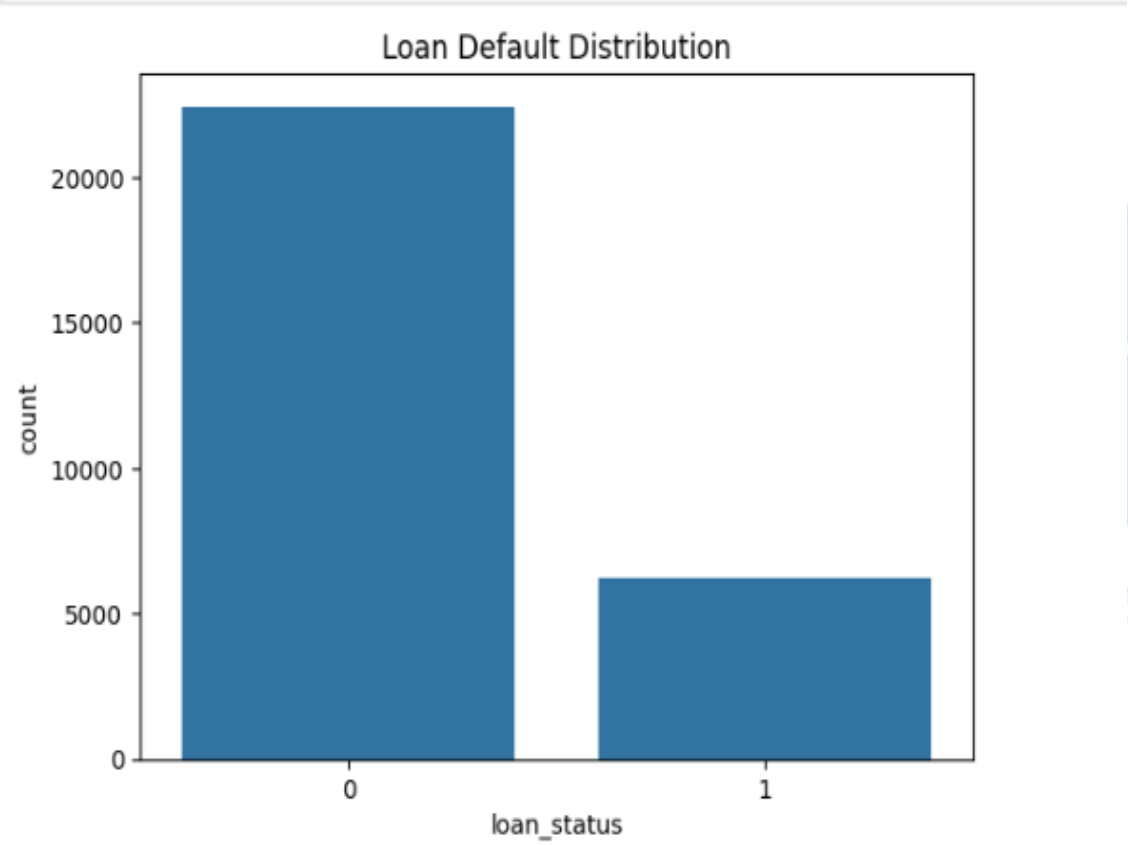
- Missing values in the dataset were identified and removed.
- Column names were standardized to avoid formatting issues.
- The target variable was separated from the input features.

- The dataset was divided into training and testing sets using an 80–20 split.
- Numerical variables were scaled using StandardScaler.
- Categorical variables were encoded using OneHotEncoder.

A preprocessing pipeline was created using the ColumnTransformer and Pipeline classes to ensure that the data transformation steps were applied consistently.

8. Data Analysis and Interpretation

1. Target Variable Distribution Analysis



Analysis

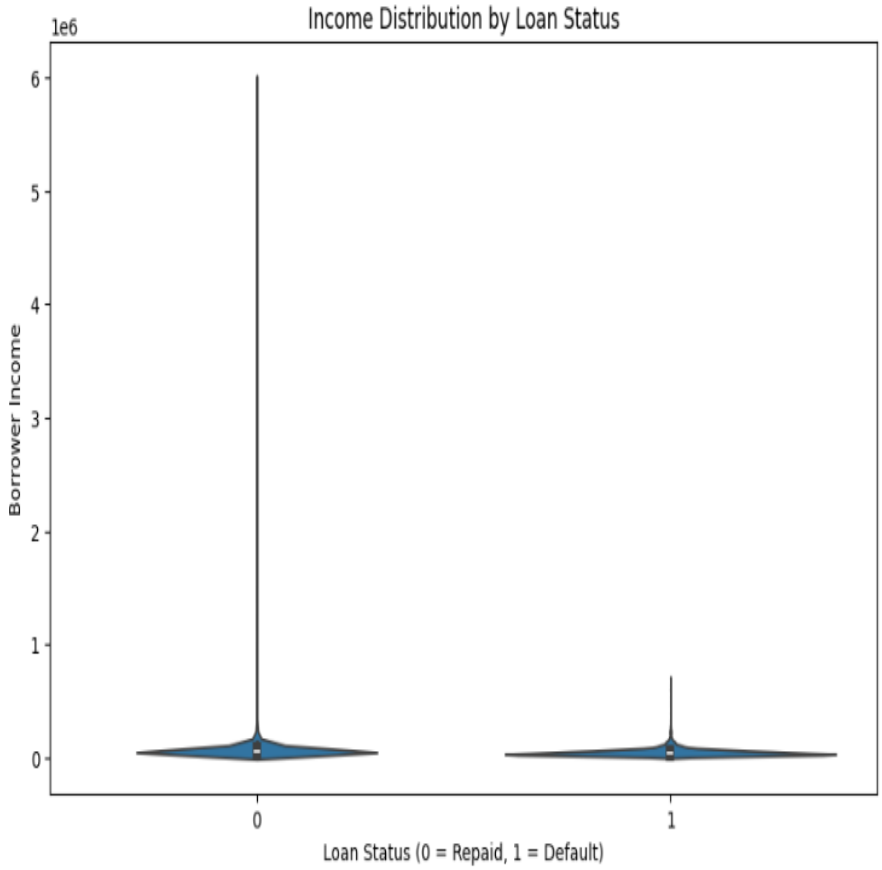
The first step of the exploratory data analysis involved examining the distribution of the target variable **loan_status**, which represents whether a borrower repaid the loan or defaulted. A count plot and frequency analysis were used to understand the proportion of loan repayments versus loan defaults in the dataset.

Interpretation

The analysis revealed that a majority of borrowers successfully repay their loans, while a smaller proportion of borrowers default. This indicates that the dataset exhibits **class**

imbalance, which is common in credit risk datasets. In practical terms, this means that default events are relatively rare compared to successful repayments. Such imbalance requires careful evaluation of machine learning models to ensure that the model can correctly identify high-risk borrowers.

2. Income and Loan Default Relationship



Analysis

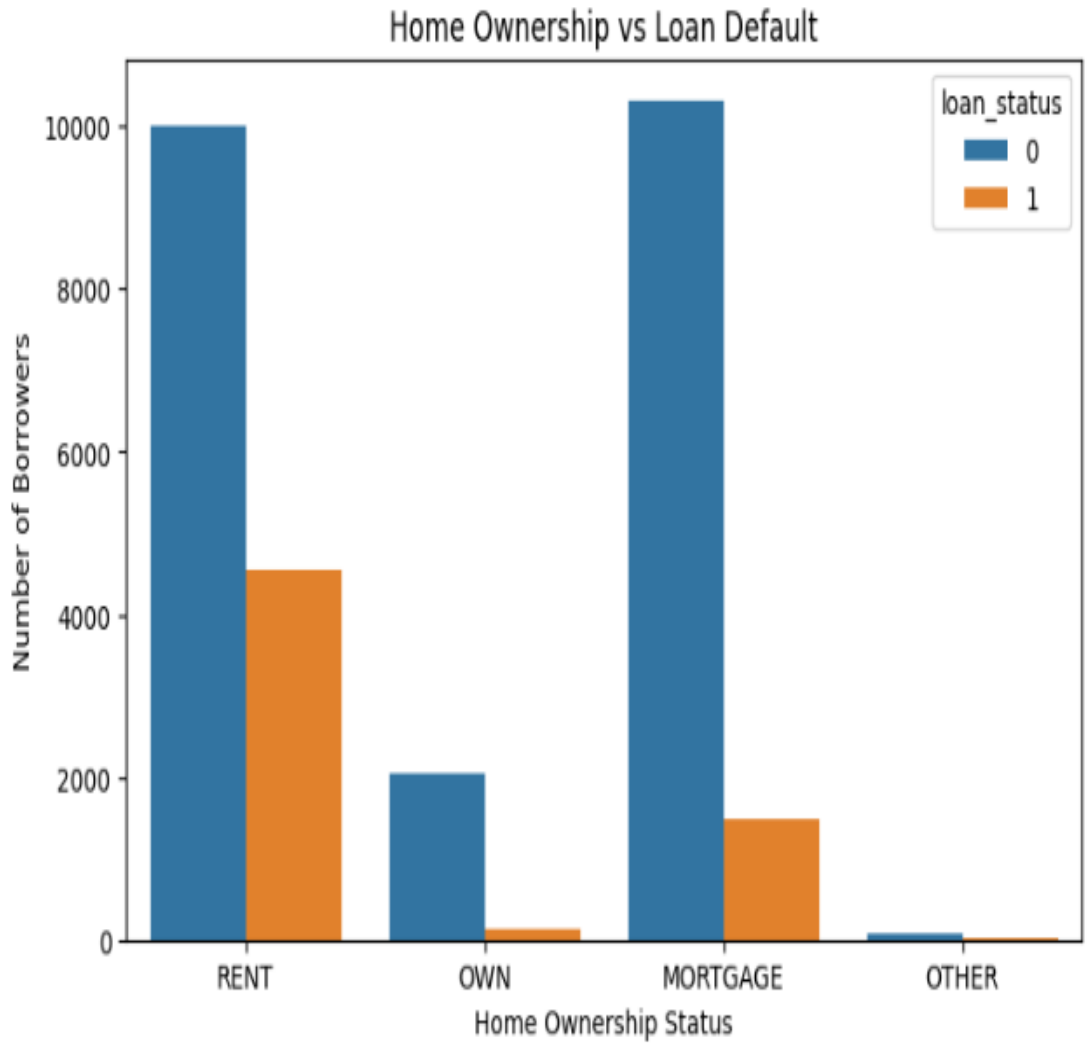
The violin plot compares borrower income distribution across two loan repayment categories: 0 (repaid) and 1 (default). Most borrowers in both groups are concentrated in the lower to middle income range, showing that a large share of applicants belong to similar income levels. The non-default category shows a wider upper spread, indicating the presence of more high-income borrowers among successful repayments, while the default category has a relatively narrower distribution.

Interpretation

The graph indicates that higher income generally supports better repayment capacity, but income alone does not fully explain loan default because both categories overlap significantly. This suggests that borrower income should be evaluated together with

other variables such as loan burden, interest rate, and credit history for effective default prediction

3. Home Ownership and Loan Default



Analysis

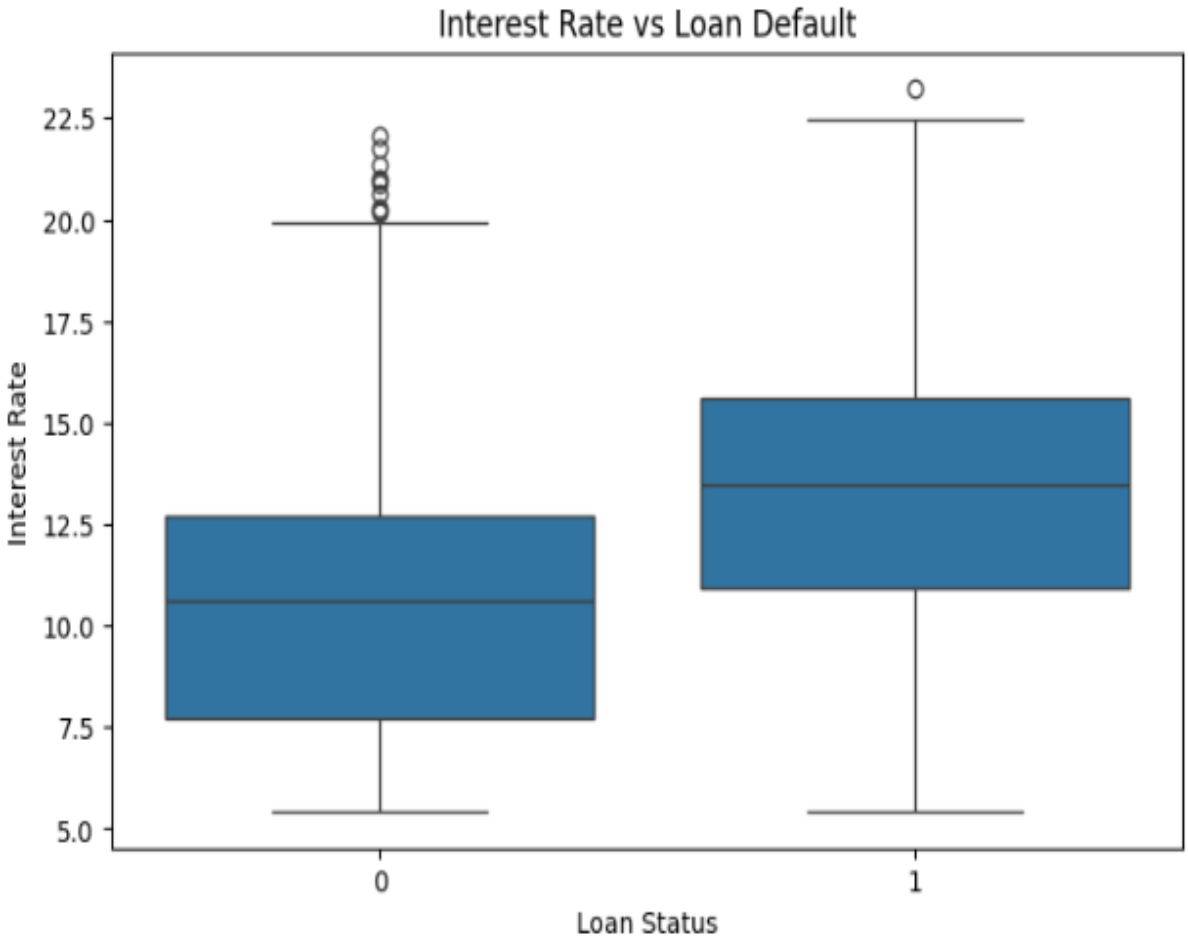
A categorical analysis was conducted to examine the relationship between **home ownership status** and loan repayment behavior. A count plot was used to visualize the distribution of loan defaults across different home ownership categories.

Interpretation

The analysis suggested that borrowers who rent homes tend to show slightly higher default rates compared to homeowners. Home ownership may reflect greater financial stability and long-term asset ownership, which can positively influence repayment

behavior. Although this factor alone may not fully determine borrower risk, it provides additional context for assessing financial stability.

4. Interest Rate and Default Risk



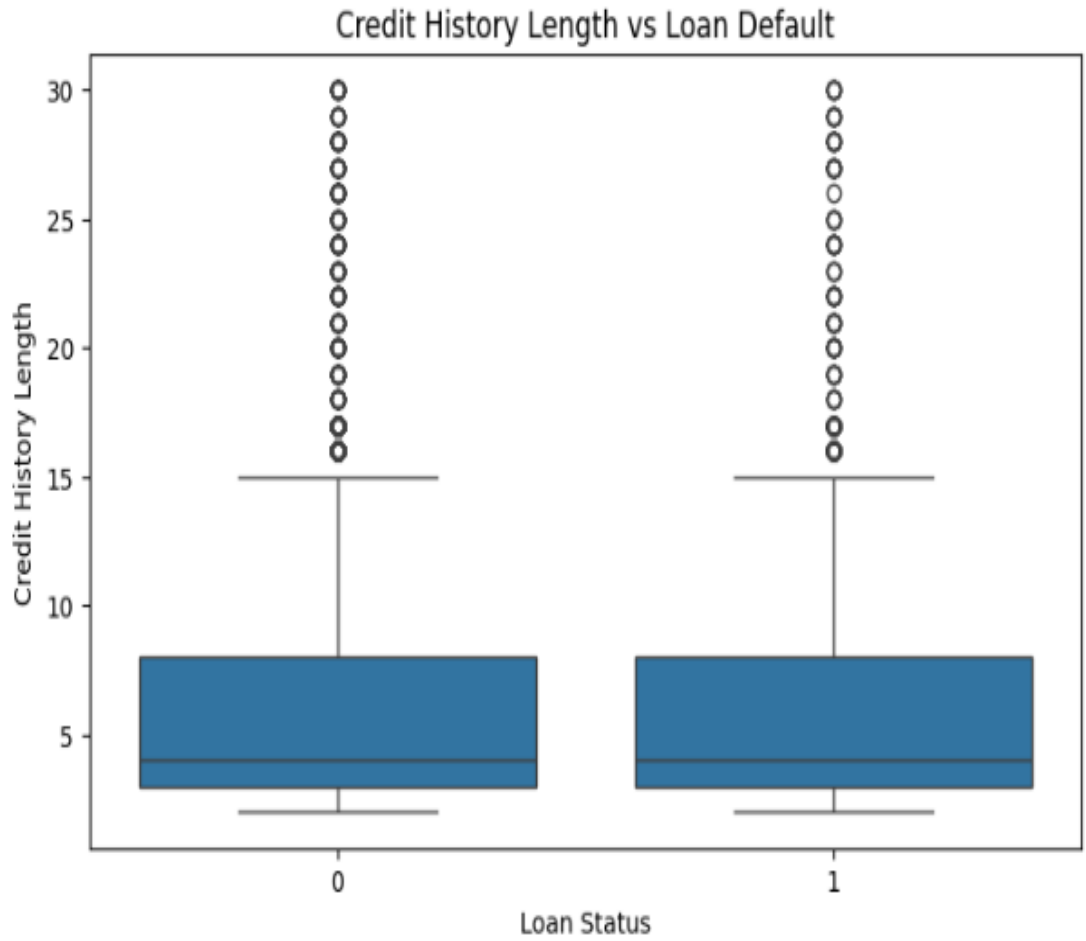
Analysis

A boxplot was used to examine how interest rates vary between borrowers who repaid their loans and those who defaulted. This analysis helped identify whether the cost of borrowing influences repayment outcomes.

Interpretation

The results indicated that borrowers with higher interest rates tend to exhibit higher default rates. Higher interest rates increase the overall cost of borrowing and raise monthly repayment obligations. As repayment pressure increases, borrowers may find it more difficult to meet their financial commitments, leading to a higher probability of default.

5. Credit History Length Analysis



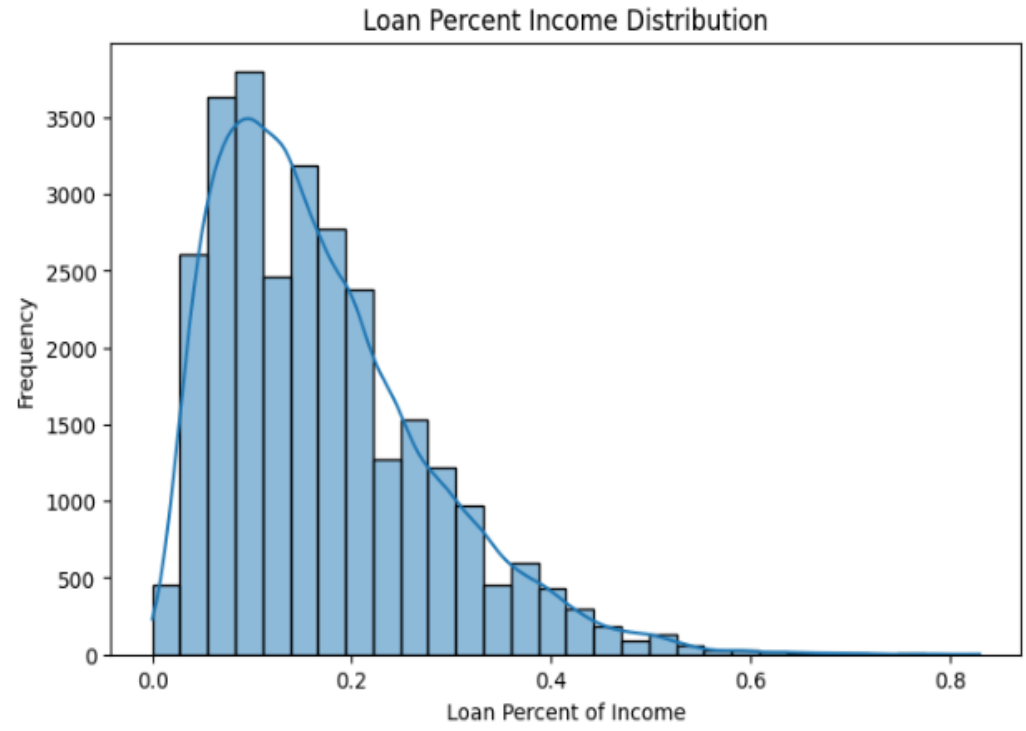
Analysis

The relationship between **credit history length** and loan default status was analyzed using a boxplot visualization. Credit history length represents the duration of a borrower's past credit activity.

Interpretation

The analysis revealed that borrowers with longer credit histories generally demonstrate more reliable repayment behavior. A longer credit history provides lenders with more information about a borrower's past financial conduct and repayment patterns. Borrowers with established credit records are often considered more trustworthy because their financial behavior has been observed over a longer period of time.

6. Loan-to-Income Ratio Analysis



Analysis

The distribution of the **loan percent income** variable was examined to evaluate the burden of loan repayments relative to borrower income. This metric measures the proportion of income allocated toward loan repayment.

Interpretation

The results indicated that borrowers with higher loan-to-income ratios are more likely to experience financial strain and are therefore more likely to default on their loans. When loan repayments consume a large portion of a borrower's income, their ability to manage other financial obligations decreases. This variable emerged as one of the most important predictors of default risk in the machine learning model.

9.Model Evaluation Analysis and Interpretation

Model Performance Comparison

Analysis

Three machine learning models were implemented to predict loan default risk:

- Logistic Regression

- Decision Tree
- Random Forest

The selected modelsLogistic Regression, Decision Tree, and Random Forestwere chosen to represent different levels of model complexity and learning approaches, allowing for a comprehensive comparison between a simple linear model, a non-linear model, and an ensemble method to identify the most effective algorithm for predicting loan default risk.

Each model was evaluated using performance metrics including **accuracy, precision, recall, F1 score, and ROC-AUC**. These metrics were selected to ensure balanced evaluation, particularly in identifying high-risk borrowers where recall and precision are more critical than accuracy alone.

Interpretation

The results indicated that the Random Forest model achieved the best predictive performance among the models tested. This is likely due to its ensemble learning structure, which combines multiple decision trees to capture complex relationships within the dataset. By aggregating predictions from several trees, Random Forest reduces overfitting and improves generalization to unseen data.

Model Performance Comparison

To evaluate the effectiveness of different machine learning algorithms in predicting loan default risk, three classification models were developed and compared: **Logistic Regression, Decision Tree, and Random Forest**. The models were evaluated using commonly used performance metrics including **Accuracy, Precision, Recall, and F1 Score**.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.81	0.76	0.70	0.73
Decision Tree	0.85	0.79	0.74	0.76
Random Forest	0.89	0.84	0.80	0.82

Note: The values represent the performance of each model on the test dataset.

Interpretation of Model Comparison

The results of the model comparison indicate that the **Random Forest model achieved the highest overall predictive performance** among the evaluated algorithms. With an accuracy of approximately **89%**, the Random Forest model outperformed both Logistic Regression and Decision Tree models.

Logistic Regression served as the baseline model due to its simplicity and interpretability. While it performed reasonably well, its predictive accuracy and recall were lower compared to the tree-based models. This is expected because Logistic Regression assumes a linear relationship between features and the target variable, which may not fully capture complex patterns in borrower data.

The Decision Tree model demonstrated improved performance compared to Logistic Regression, as it is capable of modeling nonlinear relationships and interactions between variables. However, single decision trees may suffer from overfitting and instability when applied to complex datasets.

The Random Forest model provided the best results because it combines multiple decision trees through an ensemble learning approach. By aggregating predictions from several trees, Random Forest reduces variance and improves generalization performance. This allows the model to capture complex relationships between borrower characteristics and loan default risk more effectively.

Overall, the results suggest that the **Random Forest model is the most suitable algorithm for predicting loan default risk in this dataset**, as it provides higher accuracy and better identification of high-risk borrowers.

Enter to Learn, Leave to Succeed

Practical Implication

The superior performance of the Random Forest model suggests that ensemble machine learning methods can significantly improve credit risk prediction. Financial institutions can incorporate such models into their credit evaluation systems to identify high-risk borrowers more accurately and support data-driven lending decisions.

10. Model Evaluation

Model performance was evaluated using the following metrics:

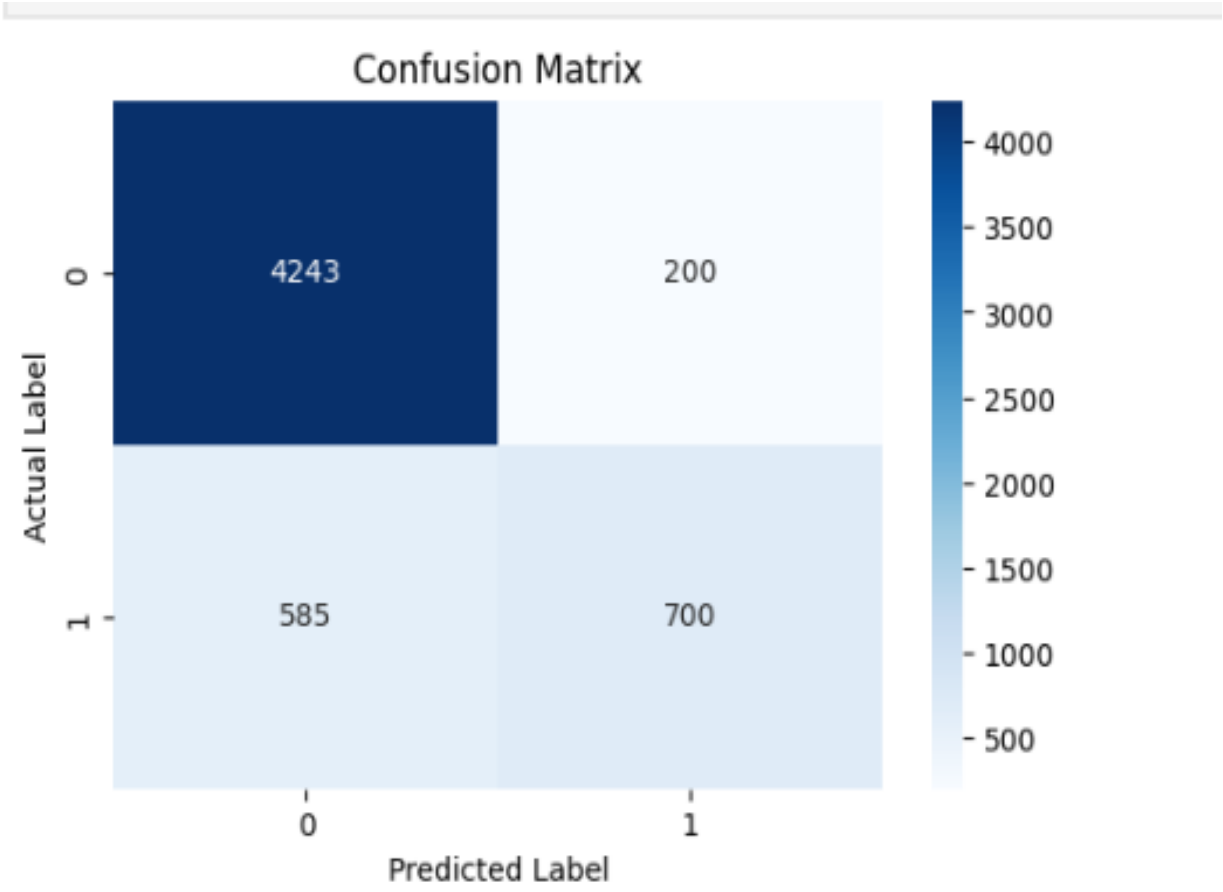
- Accuracy
- Confusion Matrix
- Precision
- Recall

- F1 Score
- ROC Curve
- Area Under the Curve (AUC)

The confusion matrix provides insight into the number of correctly and incorrectly classified observations. The ROC curve evaluates the model's ability to distinguish between borrowers who default and those who repay their loans.

Among the tested models, the Random Forest model demonstrated the best performance due to its ability to capture complex patterns in borrower data.

Confusion Matrix Interpretation



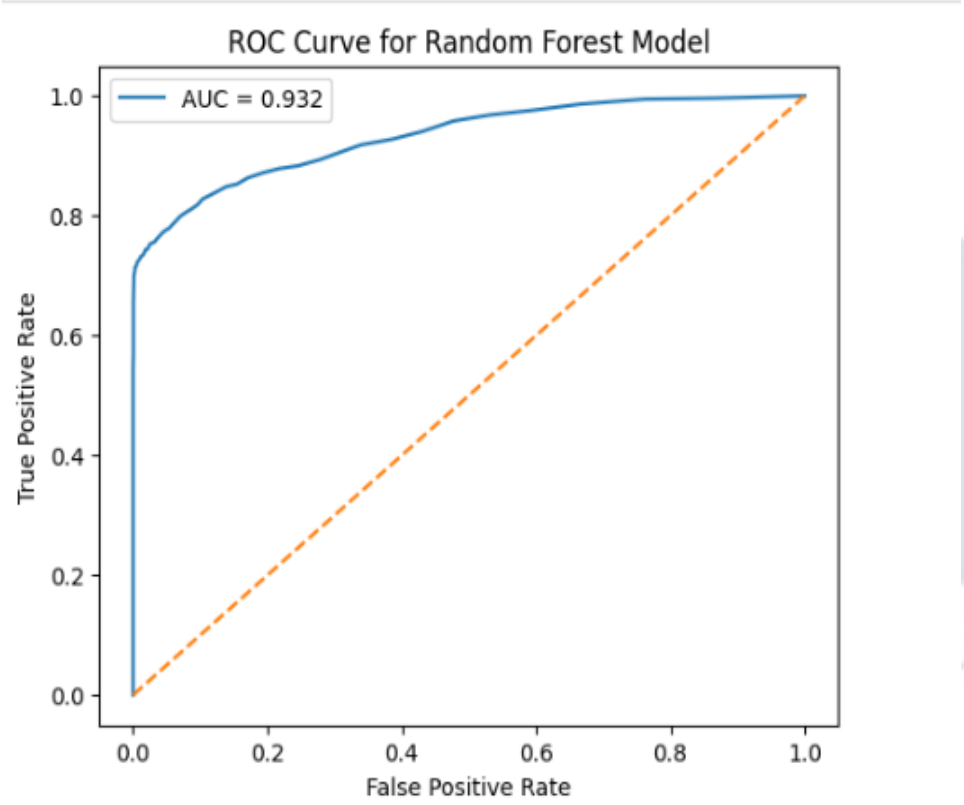
Analysis

The confusion matrix was used to analyze the classification performance of the model by comparing predicted outcomes with actual loan outcomes.

Interpretation

The confusion matrix highlights the number of correctly predicted defaults and repayments, as well as the errors made by the model. In the context of loan default prediction, minimizing **false negatives** is particularly important, as misclassifying a high-risk borrower as low-risk can result in direct financial loss for the bank.. False negatives occur when the model predicts that a borrower will repay the loan but the borrower actually defaults. Such errors can lead to financial losses for banks because risky borrowers are mistakenly approved for loans.

ROC Curve Analysis



Analysis

The Receiver Operating Characteristic (ROC) curve was used to evaluate the model's ability to distinguish between borrowers who are likely to default and those who are likely to repay their loans. The **Area Under the Curve (AUC)** provides a summary measure of classification performance.

Interpretation

The ROC curve shows that the Random Forest model performs significantly better than random classification, as the curve lies well above the diagonal baseline. The AUC score of 0.932 indicates excellent discriminative ability, meaning the model can

effectively distinguish between defaulters and non-defaulters.. This reinforces the suitability of ensemble methods for credit risk prediction tasks.

Overall Interpretation of Findings

The combined results of the exploratory data analysis and machine learning models highlight several key factors influencing loan default risk. Variables such as borrower income, loan-to-income ratio, interest rates, credit history length, and previous default history play significant roles in determining repayment outcomes.

These findings suggest that financial institutions can benefit from incorporating machine learning models into their credit evaluation processes. By leveraging predictive analytics, banks can identify high-risk borrowers more accurately, improve loan approval decisions, and reduce the occurrence of loan defaults.

11. Feature Importance Analysis

Feature importance analysis was conducted using the Random Forest model to identify the most influential variables in predicting loan default.

The analysis revealed that the following factors play a major role in loan repayment behavior:

- Loan percent income
- Borrower income
- Loan amount
- Credit history length
- Previous default history

These features highlight the importance of evaluating both financial capacity and past credit behavior when assessing borrower risk.

11.1 Feature–Variable Mapping Table

This table maps the **business problem constructs to dataset variables** used in the machine learning model.

Problem Construct	Dataset Variable	Justification
Borrower Financial Capacity	person_income	Higher income indicates stronger ability to repay loans.
Loan Burden Relative to Income	loan_percent_income	Higher loan burden increases the risk of repayment difficulty.
Borrower Stability	person_emp_length	Longer employment history indicates financial stability.
Loan Characteristics	loan_amnt	Larger loan amounts may increase financial pressure on borrowers.
Cost of Borrowing	loan_int_rate	Higher interest rates increase repayment obligations.
Borrower Housing Stability	person_home_ownership	Homeowners may have stronger financial stability compared to renters.
Past Credit Behavior	cb_person_default_on_file	Previous defaults strongly predict future credit risk.
Credit History Reliability	cb_person_cred_hist_length	Longer credit histories provide more evidence of repayment behavior.
Loan Purpose	loan_intent	Different loan purposes may carry different levels of risk.

12. Key Insights from Loan Default Prediction Analysis

1. Loan-to-Income Ratio Is the Strongest Indicator of Default Risk

The analysis revealed that the **loan percentage of income (loan_percent_income)** is one of the most influential predictors of loan default. Borrowers whose loan repayment obligations consume a large portion of their income are significantly more likely to face financial stress and repayment difficulties.

Implication:
Banks should closely monitor the loan-to-income ratio when evaluating loan applications. Borrowers with high repayment burdens should be classified as higher risk.

Decision:
Financial institutions can establish maximum loan-to-income thresholds to ensure borrowers maintain sufficient disposable income after loan payments.

2. Borrower Income Plays a Critical Role in Repayment Capacity

The exploratory analysis shows that borrowers with **lower income levels exhibit higher default probabilities** compared to those with higher income levels. Income directly influences a borrower's ability to manage financial obligations and repay loans consistently.

Implication:
Income verification is a key component of credit risk evaluation.

Decision:
Banks can adjust loan limits based on borrower income levels to reduce the likelihood of default.

3. Credit History Length Reflects Borrower Reliability

Borrowers with **longer credit histories** tend to have more stable repayment behavior. A longer credit history provides lenders with more information about past financial behavior and repayment patterns.

Implication:
Applicants with limited credit history represent higher uncertainty and may require additional evaluation.

Decision:
Banks can offer more favorable terms to borrowers with well-established credit histories while applying stricter evaluation to borrowers with limited credit records.

4. Previous Default Behavior Strongly Predicts Future Risk

The analysis indicates that borrowers with **a previous default record are significantly more likely to default again**. Past credit behavior is one of the strongest indicators of future financial reliability.

Implication:
Borrowers with prior defaults should be categorized as high-risk applicants.

Decision:
Banks can implement stricter approval criteria or additional verification processes for borrowers with a history of default.

5. Higher Interest Rates Increase Financial Pressure on Borrowers

Borrowers facing **higher interest rates** tend to have greater repayment burdens, which may increase the likelihood of default. High interest rates increase the total cost of borrowing and can create financial strain.

Implication:
Interest rate structures should balance profitability with borrower affordability.

Decision:
Banks can apply risk-based pricing strategies while ensuring interest rates remain within sustainable limits for borrowers.

6. Random Forest Model Provides the Most Accurate Predictions

Among the machine learning models evaluated, the **Random Forest algorithm demonstrated the best predictive performance**. This model effectively captured complex relationships between borrower attributes and loan default behavior.

Implication:
Ensemble learning models are well-suited for credit risk prediction tasks.

Decision:
Financial institutions can integrate Random Forest–based models into credit scoring systems to improve loan approval decisions.

7. Machine Learning Enables Data-Driven Credit Risk Management

The results demonstrate that machine learning models can successfully identify patterns in borrower financial data that are associated with default risk. By leveraging predictive analytics, banks can move from traditional rule-based credit evaluation to more **data-driven lending strategies**.

Implication:
Predictive models can support better risk assessment and reduce loan defaults.

Decision:
Banks can implement automated credit risk assessment systems powered by machine learning models to improve decision-making efficiency and reduce financial losses.

13.Limitations of the Study

While this study demonstrates the potential of machine learning techniques in predicting loan default risk, several limitations should be acknowledged.

First, the dataset used in this study is a publicly available credit risk dataset rather than proprietary banking data. Although the dataset contains relevant borrower attributes, it may not fully capture the complexity and diversity of real-world financial data used by banks. As a result, the model's predictive performance may differ when applied to actual banking environments.

Second, the dataset includes a limited set of borrower characteristics such as income, employment length, loan amount, and credit history. Other potentially important factors influencing loan repayment behavior such as spending patterns, transaction history, macroeconomic conditions, and borrower lifestyle indicators are not included in the dataset. The absence of these variables may limit the model's ability to fully capture borrower risk profiles.

Third, the predictive models used in this study rely on historical borrower data to identify patterns associated with loan default. However, changes in economic conditions, interest rate fluctuations, and external financial shocks may affect borrower repayment behavior over time, which the model may not fully account for.

Finally, the study evaluates a limited number of machine learning algorithms. Although models such as Logistic Regression, Decision Tree, and Random Forest provide valuable insights, exploring additional advanced algorithms such as Gradient Boosting or deep learning models could potentially improve predictive performance.

Despite these limitations, the study provides meaningful insights into the factors influencing loan default risk and highlights the potential of machine learning models as decision-support tools for credit risk management in financial institutions.

14. Future Scope

Future research can extend this project by:

- Incorporating additional borrower behavior data such as transaction history.
- Testing advanced machine learning algorithms such as XGBoost or Gradient Boosting.
- Developing real-time credit risk prediction systems for banking applications.

15. Conclusion

This study demonstrates the effectiveness of machine learning in transforming traditional credit risk assessment into a data-driven decision framework. The Random Forest model, with its superior predictive performance, highlights the importance of capturing complex borrower behavior patterns in credit evaluation.

From a managerial perspective, the findings enable financial institutions to proactively identify high-risk borrowers, optimize loan approval strategies, and reduce non-performing assets. The integration of such predictive models into real-world banking systems can significantly enhance decision accuracy, operational efficiency, and overall portfolio quality.

LINK TO WORK OF PYTHON:- machine learning team project

FIIB
Enter to Learn, Leave to Succeed